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Statistics in Health Technologies

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Outline

- Motivation
- Overview of current projects
- Deep dive into Howz



Motivation

- Increased collection of data
- Wide range of applications from typical "medicine" to futuristic homes.
- Sense of "doing good"
- Interesting other aspects e.g., privacy, ethics, equitability
- ...

Current Projects

- COVID-19
- Discipline Hop to Health Innovation Campus
- Seasonal emergency surgery
- QUEST with Uni Glasgow
- Howz

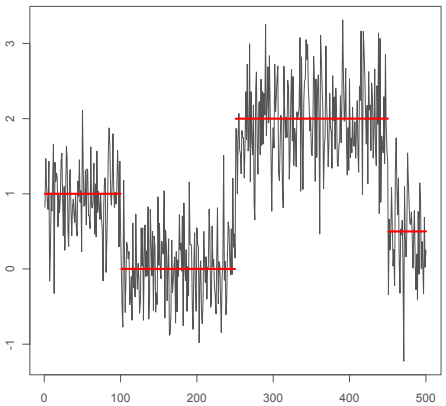
Outline

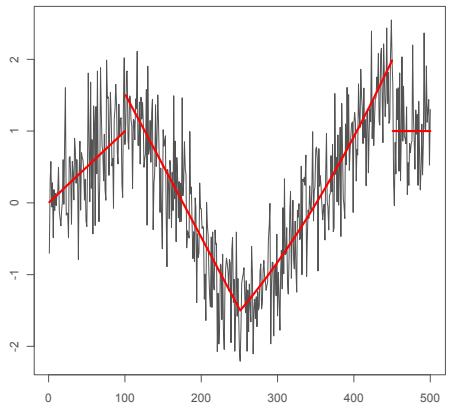
- Motivation:
- Within-period changepoint model
- RJMCMC Implementation
- Simulations
- Application

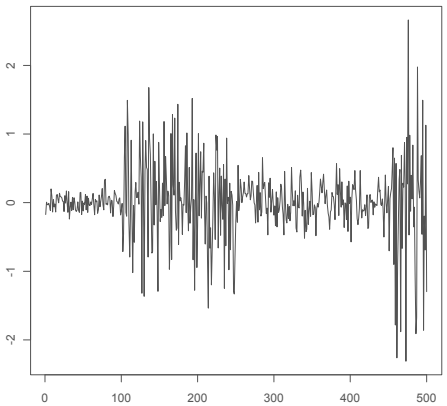


What are changepoints?

For data $y_1, \ldots, y_n$, a changepoint is a location τ where the statistical properties of $y_1, \ldots, y_\tau$ are different from $y_{\tau+1}, \ldots, y_n$ in some way.





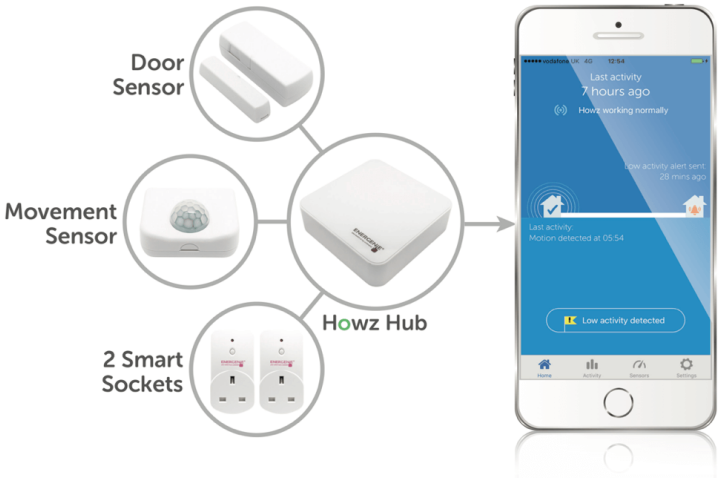


Motivation: Application

Motivation

The Howz Smart System:

- Discreet passive sensors.
- Measure activity in the home.
- Learn about your daily routine.

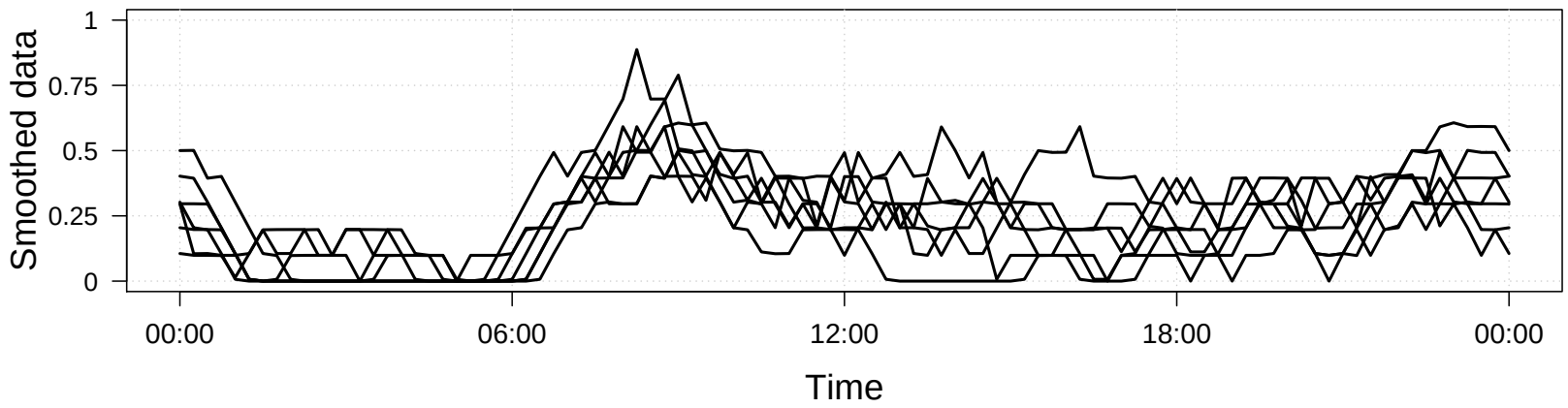


Objective:

- Identify abnormal behaviour:
  - getting up later.
  - up in the night.
  - go out less frequently.
- Indicate a decline in health or well-being in an older person.

Data

- Data frequency at 15 minute intervals (96 obs. per day).
- Binary observations: activity detected (1) or none (0).
- Interest:
  - Transition times between different activities.
  - A measure of activity intensity.
- Time series presents a daily periodic pattern.

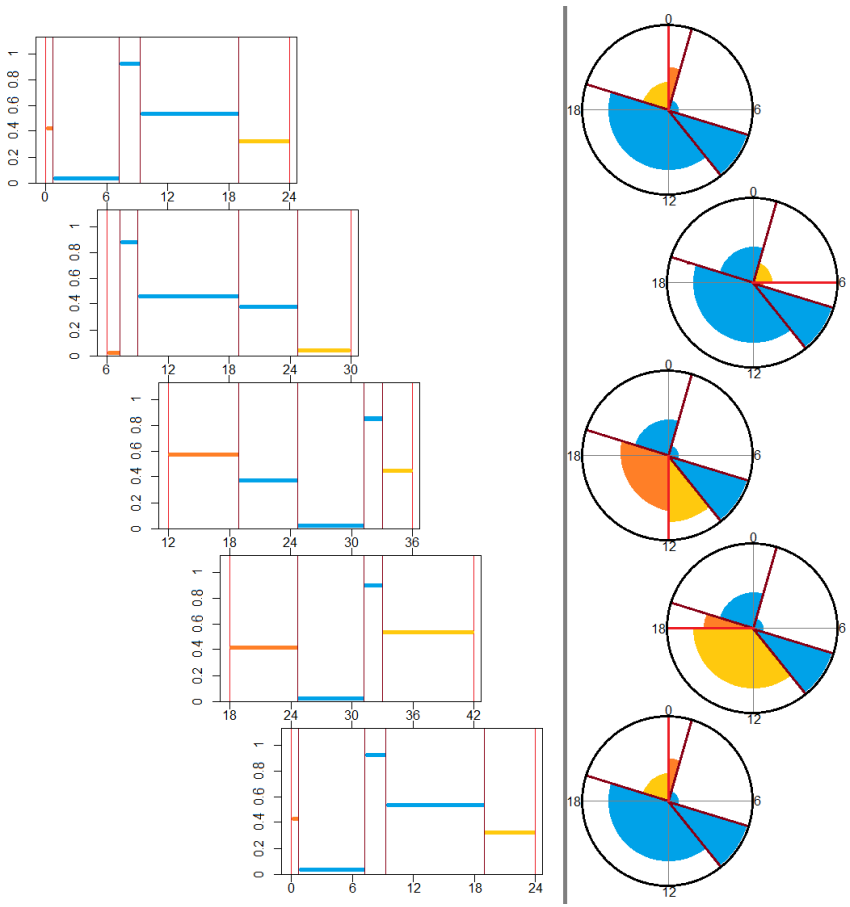


Motivation: Statistics

Changepoint Methods

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- Binary segmentation & PELT
  - Assumes conditional independence ...
  - ... but dependence exists across days.
- Multivariate time series
  - Stack days to align changepoints ...
  - ... but what should 'time zero' be.
- Circular changepoint
  - For circular data (e.g. direction) ...
  - ... but time axis is linear.
- Circular Binary Segmentation
  - local time axis is circular ...
  - ... but not truly circular perspective.
- Cyclic-HMM – Achieves periodicity via a structured transition matrix ... but period duration is stochastic.



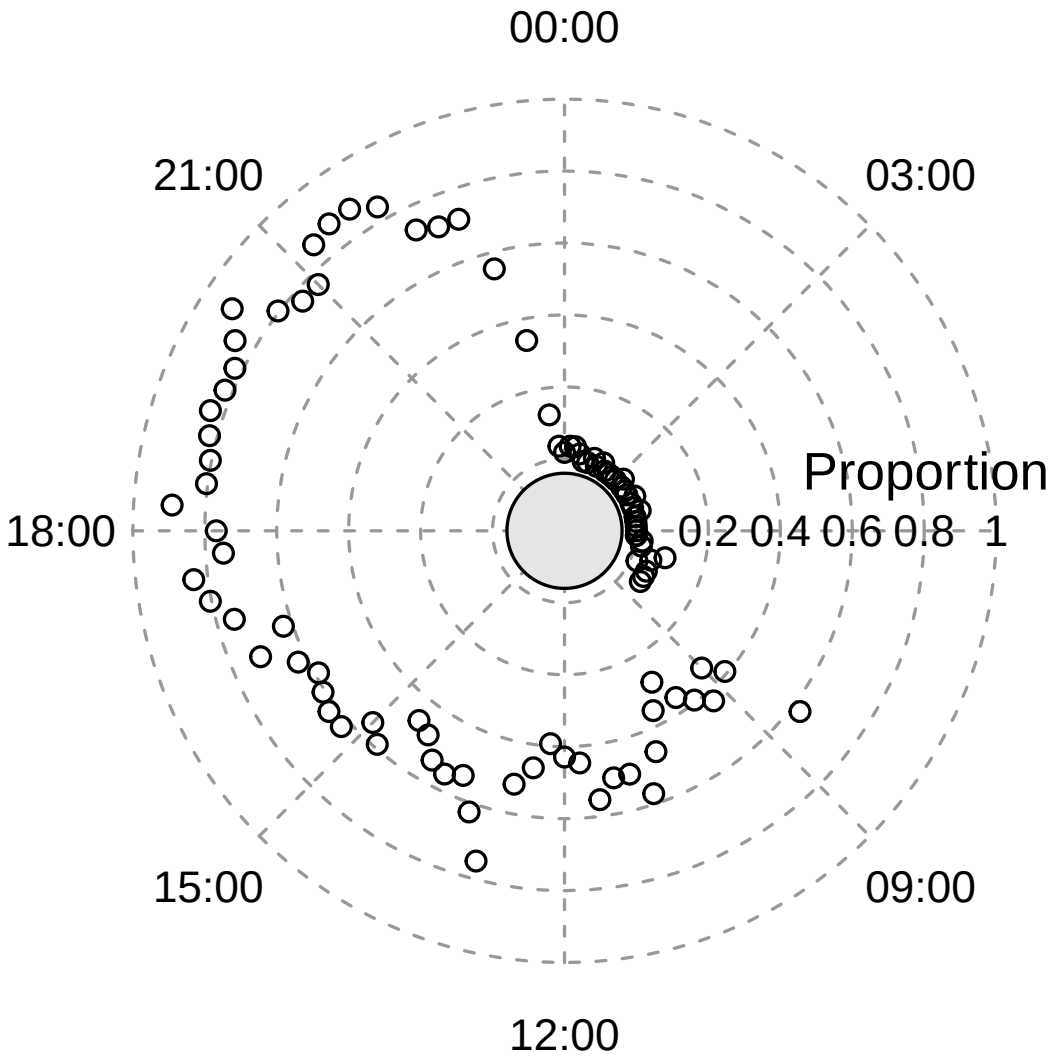
Our Contribution

Create method for within-period changepoint detection

- This talk for binary time series - have extended to real-valued data
- Bayesian method with truncated poisson prior for cpt locations
- RJMCMC for fitting
- Have extended to changes within-period and across period

Within-period changepoint detection

Data



Model

- $\{X_t\}_{t\geq 0}$ — Binary time series.
- $N \in \mathbb{N}$ — Given period length.
  - $t = cN + i$ where $c \in \mathbb{N}_0$ and $i \in \{0, 1, \ldots, N-1\}$.
- $m \geq 1$ — Number of segments partitioning the period.
- $\tau_1, \ldots, \tau_m \in \{0, 1, \ldots, N-1\}$ — Changepoint positions.
  - Ordered: $\tau_{j-1} < \tau_j$.
  - Periodic: $\tau_{m+1} = \tau_1$.
  - Minimum separation: $\tau_j - \tau_{j-1} \geq l$ for $l > 0$.
- $\phi_j \in [0, 1]$ for $j \in \{1, \ldots, m\}$ — Segment probabilities.
  - Def.: $\phi_j := \mathbb{P}(X_t = 1 | \tau_{j-1} < i \leq \tau_j)$.

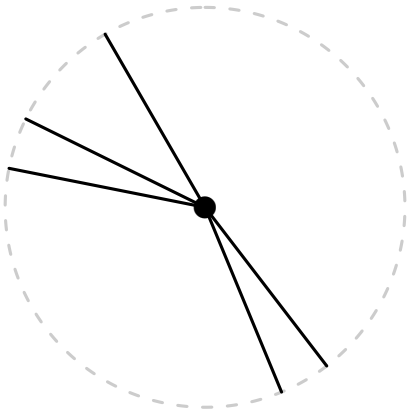
$$X_t \sim \text{Bern}\left[\phi(t)\right] \quad \text{for} \quad t \geq 0 \quad \text{where} \quad \phi(t) = \sum_{j=1}^{m} \phi_j \mathbb{I}\left\{i \in (\tau_{j-1}, \tau_j]\right\}.$$

Prior

- $m \sim \text{Truncated} - \text{Poisson}\left(1,\ \{1, \ldots, \lfloor \frac{N}{l} \rfloor\}\right)$.
- $\phi_j \sim \text{Uniform}[0,\ 1]$ for $j = 1, \ldots, m$.
- Transform $\boldsymbol{\tau}$ to excess segment length:
  - $\delta_j = \tau_j - \tau_{j-1} - l$ for $j = 1, \ldots, m$.
  - $(\delta_1, \ldots, \delta_m) \sim \text{Dirichlet} - \text{Multinomial}(N - lm,\ \alpha \mathbf{1}_m)$
- $\tau_1 \sim \text{Uniform}\{0, \ldots, N-1\}$.

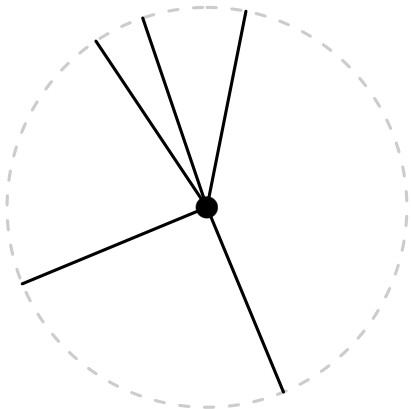
Cluster

$\alpha = 0.5$



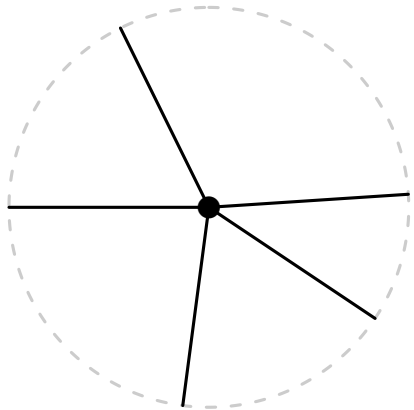
Random

$\alpha = 1$



Repel

$\alpha = 2$



Posterior

Segment probabilities:

$$\phi_j|\boldsymbol{\tau}, m, \boldsymbol{x} \sim \text{Beta}\left(1 + s_j,\ 1 + n_j - s_j\right) \quad \text{for } j = 1, \ldots, m,$$

where $n_j = \sum_t \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$ and $s_j = \sum_t x_t \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$.

Number of segments and changepoint positions:

$$\pi(\boldsymbol{\tau}, m|\boldsymbol{x}) \propto \begin{cases} B(1 + \sum_t x_t,\ 1 + \sum_t (1 - x_t)) & m = 1, \\ \prod_{j=1}^{m} \left[\frac{B(1+s_j,\ 1+n_j-s_j)}{(\alpha+\delta_j)B(\alpha,\ \delta_j+1)}\right] \frac{(m\alpha+N-lm)B(m\alpha,\ N-lm+1)}{m!N} & \text{o/w}, \end{cases}$$

where $B(\cdot, \cdot)$ is the beta function.

- Trivial to evaluate $\pi(\boldsymbol{\tau}, m|\boldsymbol{x})$ for a given changepoint vector.
- Support size is potentially impractically large:
  - For $N = 96$ and $l = 4$, then there are $\sim 27.3 \times 10^{12}$ events.

Reversible Jump Monte Carlo Markov Chain (RJMCMC)

RJMCMC

- Permits exploration of varying dimensional spaces.
- Need to satisfy 'dimension-matching' condition.
- Focus inference on the marginal posterior $\pi(\boldsymbol{\tau}, m|\boldsymbol{x})$.

Algorithm

1. Sample a segment $j \in \{1, \ldots, m\}$ uniformally.
2. Define the proposal set $\mathcal{P}_j(\boldsymbol{\tau}, m)$.
3. Sample a changepoint vector from the proposal pmf:

$$q(\boldsymbol{\tau}', m'|\boldsymbol{\tau}, m) \propto \pi(\boldsymbol{\tau}', m'|\boldsymbol{x}) \mathbb{I}\left\{(\boldsymbol{\tau}', m') \in \mathcal{P}_j(\boldsymbol{\tau}, m)\right\}.$$

4. Accept/Reject proposal. Return to 1.

- Evaluate $\pi(\boldsymbol{\phi}|\boldsymbol{\tau}, m, \boldsymbol{x})$ given the changepoint posterior samples.

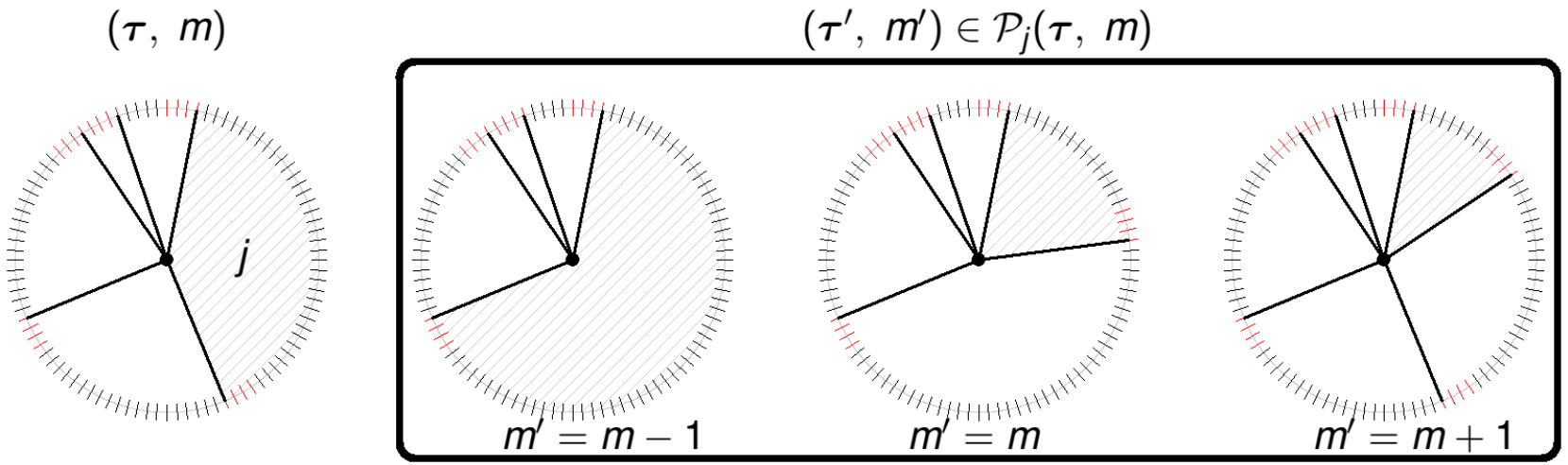
Proposal Set

Proposal set, $\mathcal{P}_j(\tau, m)$, contains three move types:

- Combine sampled segment with the next.
- Contract or extend the length of the current segment.
- Partition current segment into two.

Unique mapping

For each $(\tau', m') \in \mathcal{P}_j(\tau, m)$ excluding (τ, m), there exists a unique segment j' such that $(\tau, m) \in \mathcal{P}_{j'}(\tau', m')$.



Simulations

Monte Carlo Error

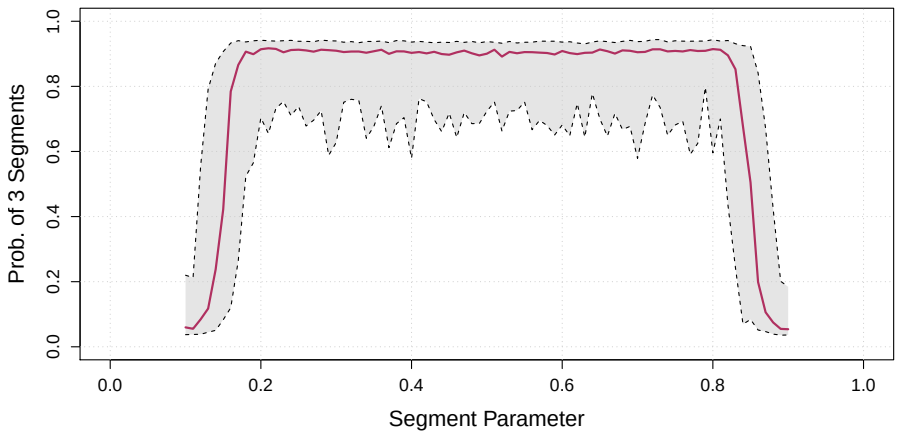
- $N = 24, I = 4, \tau = (0, 8, 16), \phi = (0.25, 0.5, 0.6), \gamma = 1, 30$ cycles

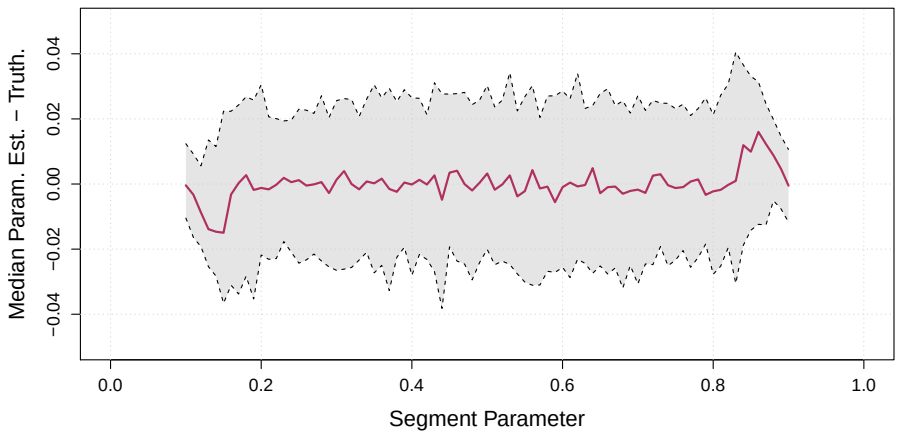
Event Positions	True Probability	Bias ($\times 10^{-3}$)	RMSE ($\times 10^{-3}$)
0, 9, 16	0.316	-0.062	7.024
0, 8, 16	0.261	0.157	6.720
0, 16	0.150	0.106	5.503
0, 7, 16	0.048	-0.020	3.153
1, 9, 16	0.023	0.014	1.949
1, 8, 16	0.019	-0.101	1.658

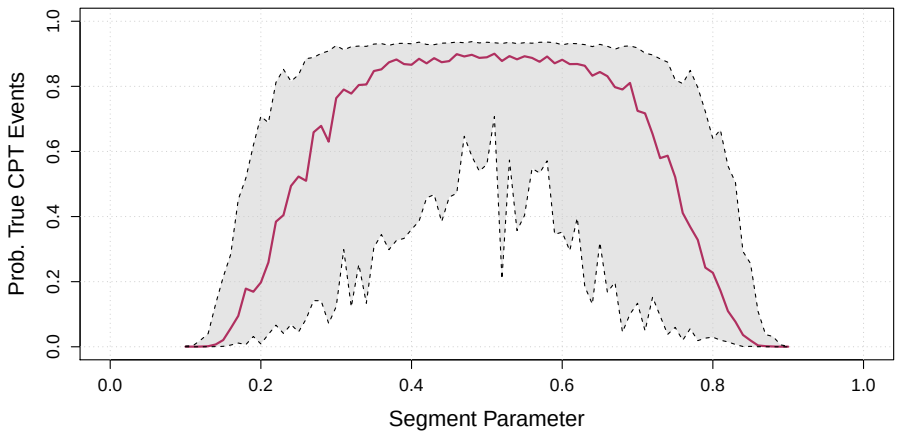
Simulation

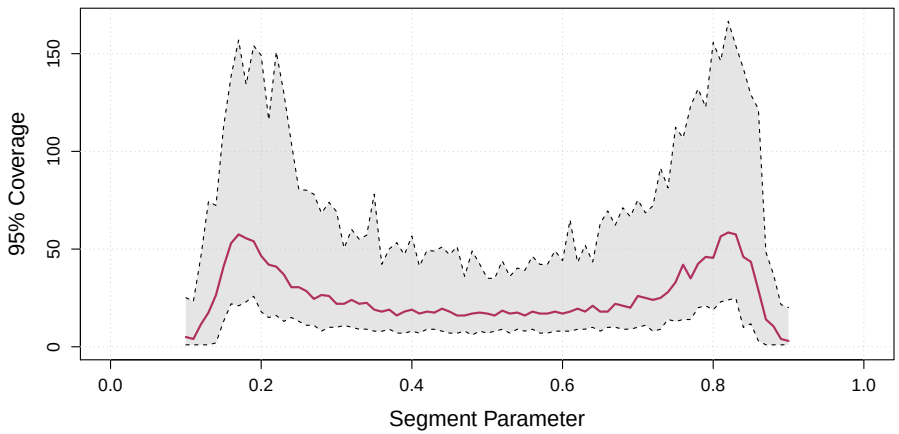
- Period length: $N = 96$ – 15 minute frequency.
- Number of periods: $c = 28$ – 4 weeks.
- CPT events: $\boldsymbol{\tau}_0 = (0, 32, 64)$ – evenly spread. $(m_0 = 3)$
- Segment parameter: $\boldsymbol{\phi}_0 = (0.1, \varphi, 0.9)$ for $\varphi \in [0.1, 0.9]$.
- Generated 100 data sets under each scenario.
- Minimum segment length: $l = 4$ (1 hour)
- Initiated two chains:
  - $\boldsymbol{\tau}_{ch1}^{(0)} = \emptyset$ – single segment model.
  - $\boldsymbol{\tau}_{ch2}^{(0)} = (0,\ l,\ 2l, \ldots,\ N - 2l,\ N - l)$ – saturated model.
- Draw 10,000 samples from the posterior after convergence.

Simulation



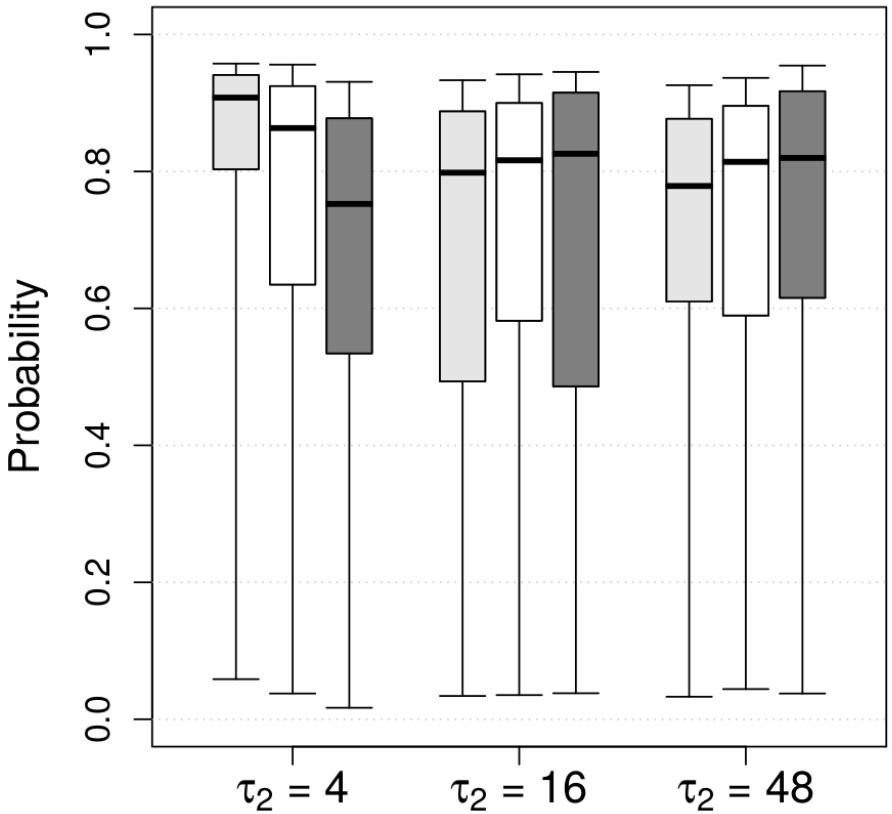


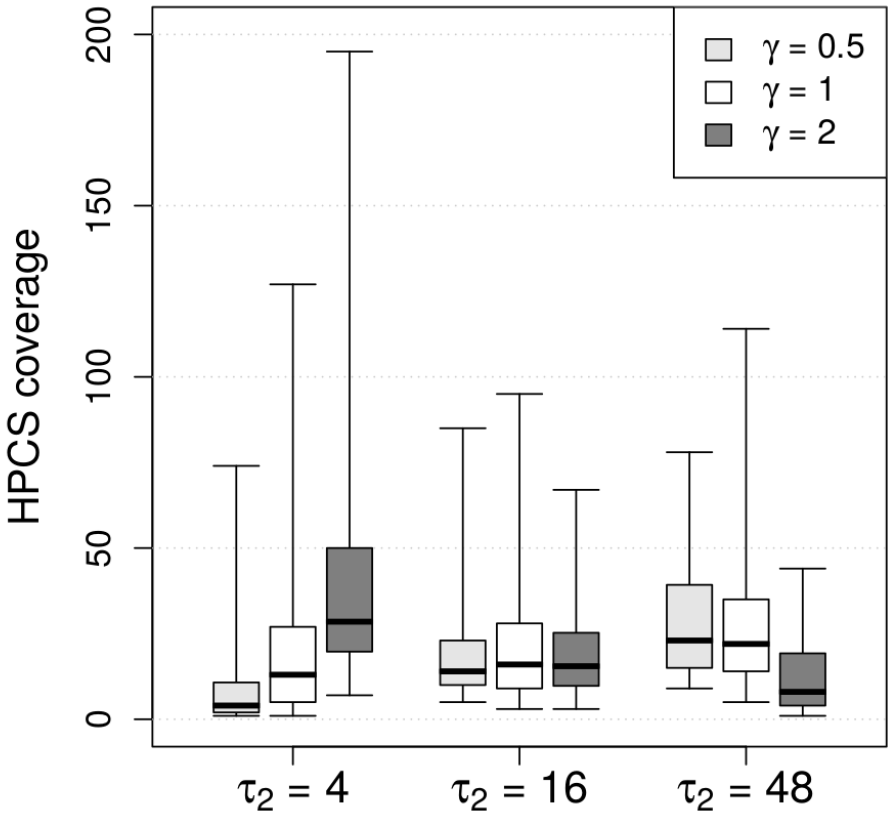




Median estimates (red) and 90% Monte Carlo interval (grey).
TL: $\hat{\pi}(m_0|\boldsymbol{x})$, TR: $\varphi_{med.} - \varphi_0$, BL: $\hat{\pi}(\boldsymbol{\tau}_0|\boldsymbol{x})$, BR: 95% Posterior coverage.

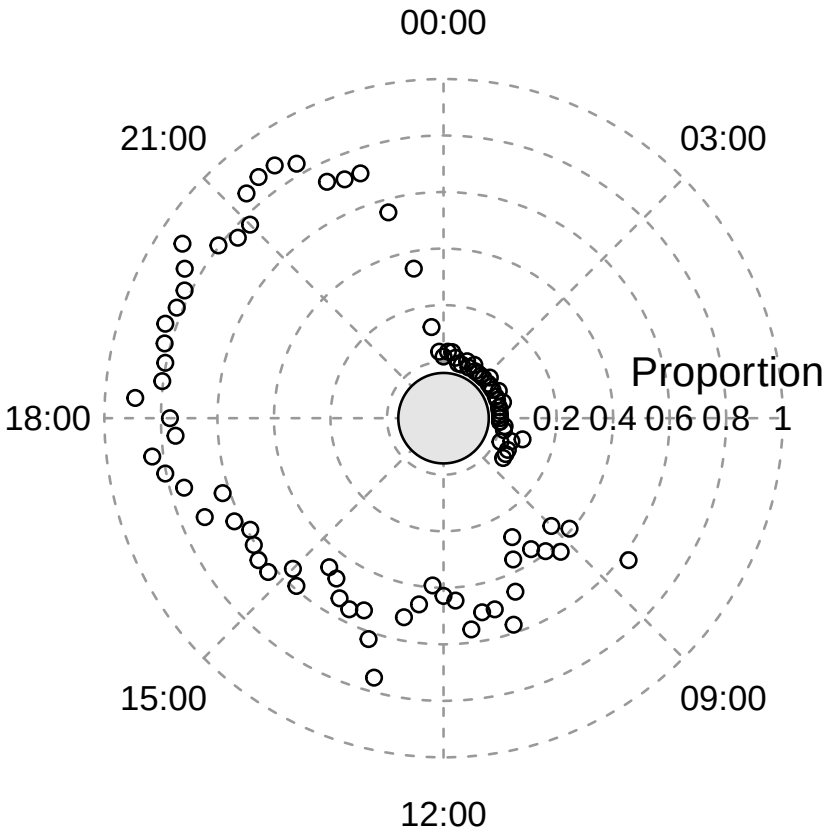
Prior Sensitivity





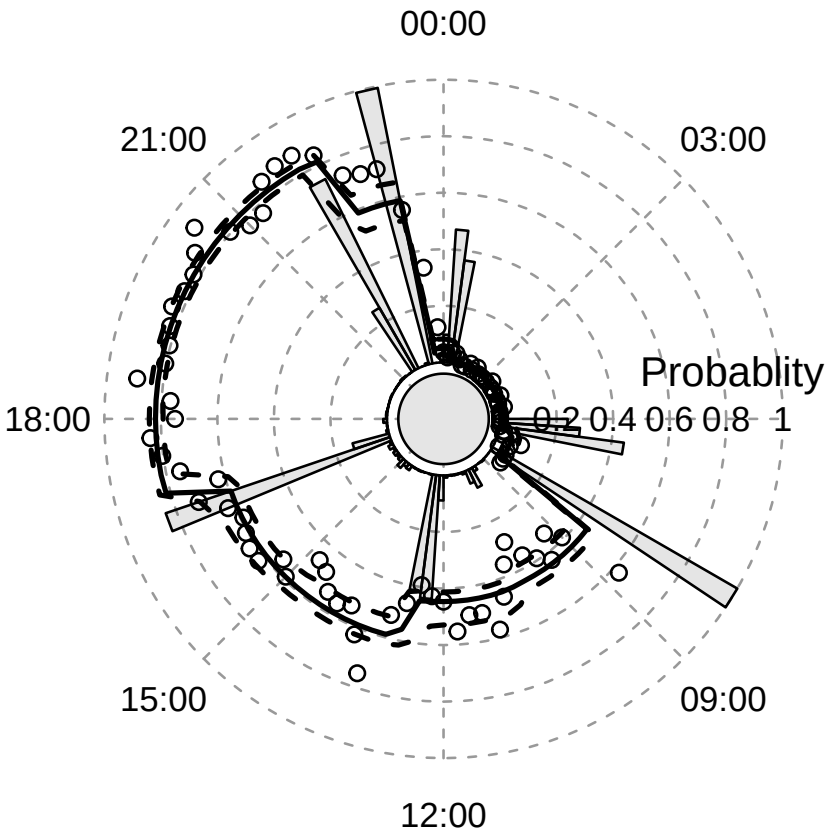
Application

Case Study



- 8 weeks (Summer 2018)
- 15 minute intervals (N=96, n=5376)
- Assume $l = 4$ (1 hour)
- Run two chains to validate convergence.
- Draw 10^4 samples from the posterior.

Case Study - Results



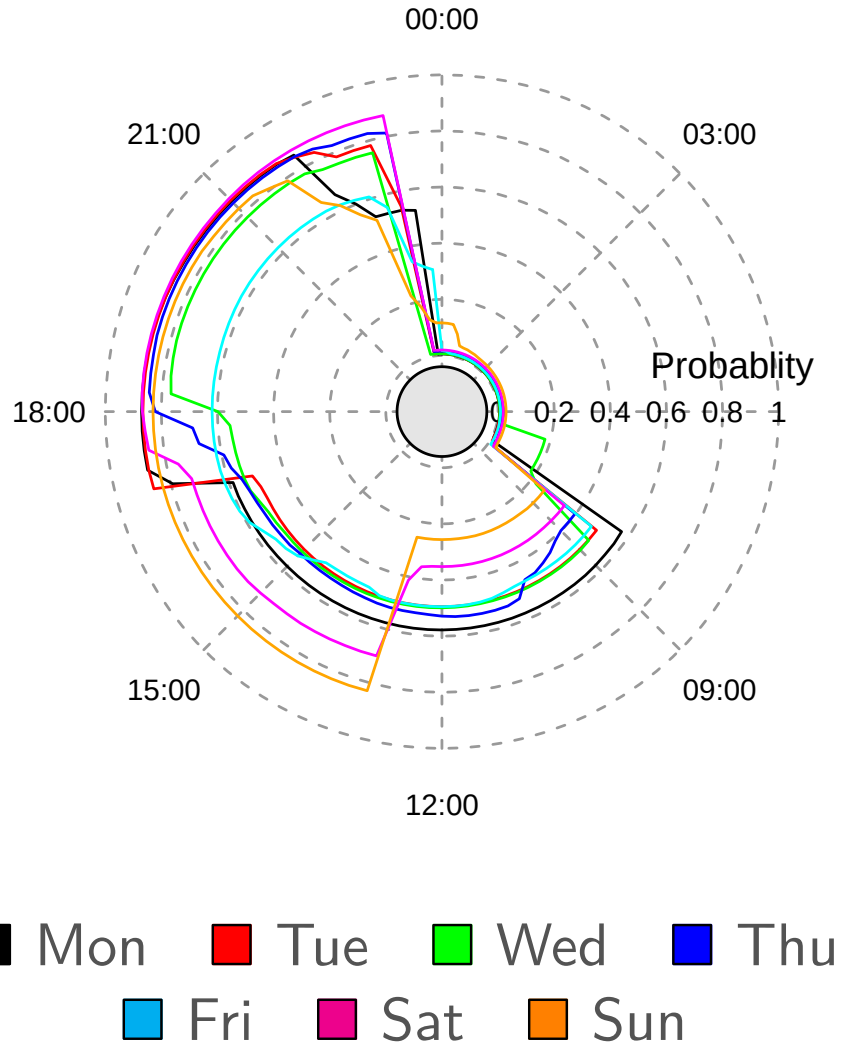
Modal number of segments:
$\hat{m} = 7$, $\hat{\pi}(\hat{m}|\boldsymbol{x}) = 75.1\%$

Modal changepoint vector, $\hat{\boldsymbol{\tau}}$:

Time	$\hat{\pi}(i \in \boldsymbol{\tau}\|\boldsymbol{x})$	$\hat{\phi}_j$
00:30	47.1%	5.0%
06:45	44.3%	0.6%
08:15	100.0%	5.4%
12:45	41.5%	44.5%
16:45	83.8%	59.4%
22:15	74.6%	81.8%
23:15	99.4%	58.0%

$\hat{\pi}(\hat{\boldsymbol{\tau}}|\boldsymbol{x}) = 5.2\%$
80% coverage: 268 events

Case Study - Weekday



- Few events estimated:
  $\hat{m} = 4$ or 5.
- Consistent evening and night activity levels.
- Small variation in night and morning events.
- Weekday dependence on afternoon-evening transition:
  Weekday: 16:45–18:00.
  Weekend: 12:45.

Discussion

Overview:

- A new approach for identifying within-period changepoint events for binary time series.

  Taylor, S. A. C., Killick, R., Burr, J. and Rogerson, L. (2018)

  'Changepoint Detection within Periodic Binary Time Series'. Under Review.

- Pooling of evidence across multiple periods by applying a circular perspective to time.

Next steps:

- Periodic changepoint patterns of non-binary time series.
- Investigate weekday effects.